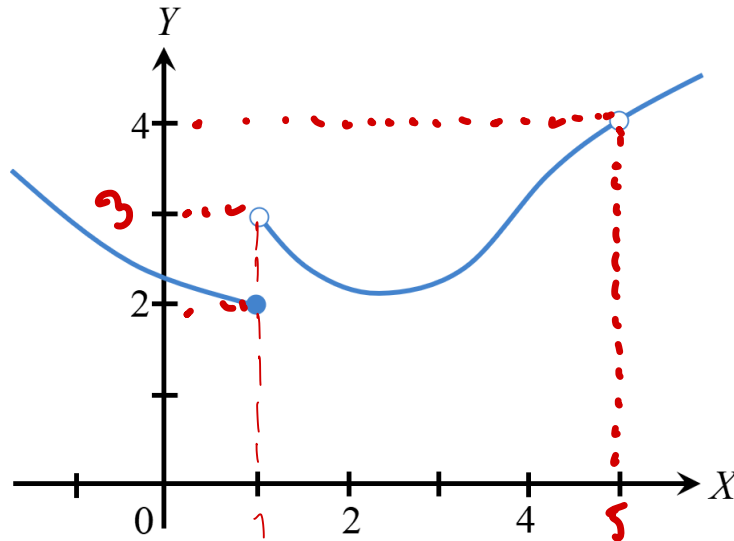


คาบทบทวน CALCULUS I สัปดาห์ที่ 1

1. กำหนดกราฟของฟังก์ชัน f ดังรูป



จงหา

(a) $f(1) = 2$

(e) $f(5)$ ไม่มีค่า

(b) $\lim_{x \rightarrow 1^-} f(x) = 2$

(f) $\lim_{x \rightarrow 5^-} f(x) = 4$

(c) $\lim_{x \rightarrow 1^+} f(x) = 3$

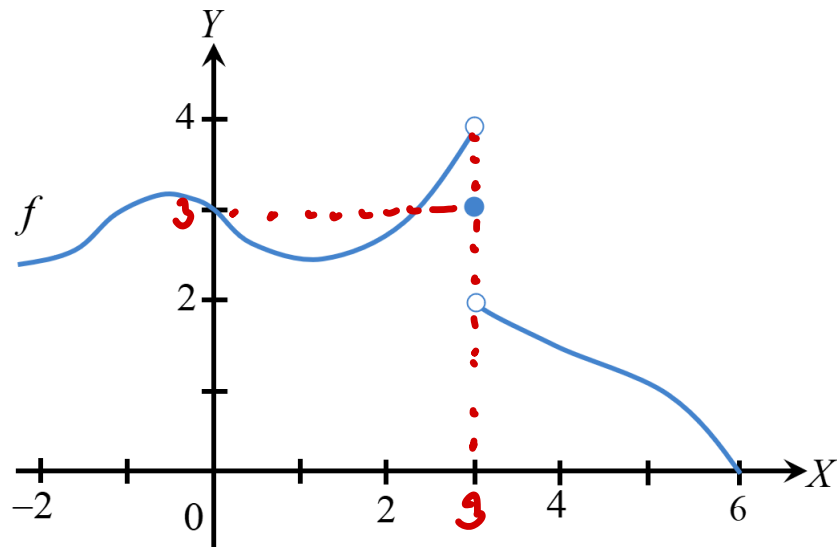
(g) $\lim_{x \rightarrow 5^+} f(x) = 4$

(d) $\lim_{x \rightarrow 1} f(x)$ หาไม่ได้

(h) $\lim_{x \rightarrow 5} f(x) = 4$

เพราะ: ลิมิตทางขวา กับ ลิมิตทางซ้าย ไม่เท่ากัน

2. กำหนดกราฟของฟังก์ชัน f ดังรูป



จงหา

(a) $f(0) = 3$

(d) $\lim_{x \rightarrow 3^-} f(x) = 4$

(b) $\lim_{x \rightarrow 0} f(x) = 3$

(e) $\lim_{x \rightarrow 3^+} f(x) = 2$

(c) $f(3) = 3$

(f) $\lim_{x \rightarrow 3} f(x)$ **หาค่าไม่ได้**

3. จงหาลิมิตต่อไปนี้ ถ้าลิมิตมีค่า

(a) $\lim_{x \rightarrow 0} (3x^2 + 7x - 12)$

$$= 3(0)^2 + 7(0) - 12$$

$$= 0 + 0 - 12 = -12 \quad \#$$

(b) $\lim_{x \rightarrow -1} (x^5 - 2x)$

$$= (-1)^5 - 2(-1)$$

$$= -1 - (-2)$$

$$= -1 + 2 = 1 \quad \#$$

$$(c) \lim_{x \rightarrow 2} x^5(x-2)$$

$$= 2^5(2-2)$$

$$= 2^5(0)$$

$$= 0 \quad \#$$

$$(d) \lim_{x \rightarrow -1} (x+3)(x^2+2)$$

$$= (-1+3)((-1)^2+2)$$

$$= (2)(1+2)$$

$$= (2)(3) = 6 \quad \#$$

$$(e) \lim_{x \rightarrow 3} \frac{x+1}{2x-5}$$

$$= \frac{3+1}{2(3)-5} = \frac{4}{6-5} = \frac{4}{1} = 4 \quad \#$$

$$(f) \lim_{x \rightarrow -5} \frac{x^2-25}{x+5} = \frac{(-5)^2-25}{-5+5} = \frac{25-25}{0} = \frac{0}{0}$$

$$x^2 - a^2 = (x-a)(x+a)$$

$$\lim_{x \rightarrow -5} \frac{x^2-25}{x+5} = \lim_{x \rightarrow -5} \frac{x^2-5^2}{x+5} = \frac{(x-5)(x+5)}{(x+5)}$$

$$= x-5 = -5-5 = -10 \quad \#$$

$$(g) \lim_{x \rightarrow -1} \frac{x+1}{x^2-x-2} = \lim_{x \rightarrow -1} \frac{\cancel{x+1}}{(x-2)(\cancel{x+1})}$$

$$= \lim_{x \rightarrow -1} \frac{1}{x-2}$$

$$= \frac{1}{-1-2} = \frac{1}{-3} = -\frac{1}{3} \#$$

$$(h) \lim_{x \rightarrow -1} \frac{x^2-x-2}{x^2+4x+3} = \frac{(-1)^2-(-1)-2}{(-1)^2+(-1)4+3} = \frac{0}{0}$$

$$\lim_{x \rightarrow -1} \frac{(x-2)(\cancel{x+1})}{(x+3)(\cancel{x+1})} = \lim_{x \rightarrow -1} \frac{(x-2)}{(x+3)} = \frac{-1-2}{-1+3} = \frac{-3}{2} \#$$

$$(i) \lim_{x \rightarrow 1} \frac{1-\sqrt{x}}{1-x}$$

$$\tilde{u} - \tilde{v} = (u-a)(u+a)$$

$$= \lim_{x \rightarrow 1} \frac{1-\sqrt{x}}{1-x} \times \frac{1+\sqrt{x}}{1+\sqrt{x}}$$

$$= \lim_{x \rightarrow 1} \frac{(1-\sqrt{x})(1+\sqrt{x})}{(1-x)(1+\sqrt{x})}$$

$$= \lim_{x \rightarrow 1} \frac{(1\cancel{-x})}{(1\cancel{-x})(1+\sqrt{x})}$$

$$= \lim_{x \rightarrow 1} \frac{1}{1+\sqrt{x}} = \frac{1}{1+\sqrt{1}} = \frac{1}{2} \#$$

$$(j) \lim_{x \rightarrow 9} \frac{3 - \sqrt{x}}{9 - x}$$

$$\lim_{x \rightarrow 9} \frac{(3 - \sqrt{x})}{9 - x} \cdot \frac{(3 + \sqrt{x})}{3 + \sqrt{x}}$$

$$\lim_{x \rightarrow 9} \frac{(9 - x)}{(9 - x)(3 + \sqrt{x})} = \lim_{x \rightarrow 9} \frac{1}{3 + \sqrt{x}} = \frac{1}{3 + \sqrt{9}} = \frac{1}{3 + 3}$$

$$= \frac{1}{6} \#$$

$$(k) \lim_{x \rightarrow 1} \frac{2 - \sqrt{x+3}}{x-1}$$

$$= \lim_{x \rightarrow 1} \frac{(2 - \sqrt{x+3})}{x-1} \cdot \frac{(2 + \sqrt{x+3})}{2 + \sqrt{x+3}}$$

$$1 - x = -(-1 + x)$$

$$= \lim_{x \rightarrow 1} \frac{4 - (x+3)}{(x-1)(2 + \sqrt{x+3})} = \lim_{x \rightarrow 1} \frac{4 - x - 3}{(x-1)(2 + \sqrt{x+3})} = \lim_{x \rightarrow 1} \frac{1-x}{(x-1)(2 + \sqrt{x+3})} = \lim_{x \rightarrow 1} \frac{-(-1+x)}{(x-1)(2 + \sqrt{x+3})}$$

$$= \lim_{x \rightarrow 1} \frac{-(x-1)}{(x-1)(2 + \sqrt{x+3})}$$

$$= \lim_{x \rightarrow 1} \frac{-1}{2 + \sqrt{x+3}} = \frac{-1}{2 + \sqrt{1+3}} = \frac{-1}{2+2} = -\frac{1}{4} \#$$

$$(l) \lim_{x \rightarrow 0} \frac{\sqrt{x^2 + 100} - 10}{x^2}$$

$$= \lim_{x \rightarrow 0} \frac{\sqrt{x^2 + 100} - 10}{x^2} \cdot \frac{\sqrt{x^2 + 100} + 10}{\sqrt{x^2 + 100} + 10}$$

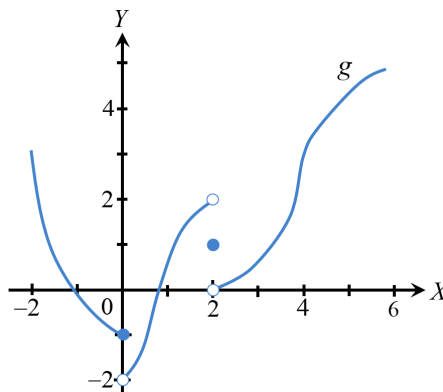
$$= \lim_{x \rightarrow 0} \frac{(x^2 + 100) - (100)}{x^2 \cdot (\sqrt{x^2 + 100} + 10)} = \lim_{x \rightarrow 0} \frac{x^2}{x^2 (\sqrt{x^2 + 100} + 10)}$$

$$= \lim_{x \rightarrow 0} \frac{1}{\sqrt{x^2 + 100} + 10}$$

$$= \frac{1}{10 + 10} = \frac{1}{20} \#$$

แบบฝึกหัดเพิ่มเติม

1. กำหนดกราฟของฟังก์ชัน g ดังรูป



จงหา

- | | | |
|-------------------------------------|-------------------------------------|-------------------------------------|
| (a) $g(0)$ | (d) $\lim_{x \rightarrow 0} g(x)$ | (g) $\lim_{x \rightarrow 2^+} g(x)$ |
| (b) $\lim_{x \rightarrow 0^-} g(x)$ | (e) $g(2)$ | (h) $\lim_{x \rightarrow 2} g(x)$ |
| (c) $\lim_{x \rightarrow 0^+} g(x)$ | (f) $\lim_{x \rightarrow 2^-} g(x)$ | (i) $\lim_{x \rightarrow 4} g(x)$ |

2. จงหาขีดจำกัดต่อไปนี้ ถ้าขีดจำกัดมีค่า

- | | |
|---|--|
| (a) $\lim_{x \rightarrow 5} (2x^2 + 3x - 1)$ | (k) $\lim_{x \rightarrow 1} \frac{\sqrt{x^2 - 1}}{x - 1}$ |
| (b) $\lim_{x \rightarrow 2} (x^3 + 4x^2 - 3)$ | (l) $\lim_{u \rightarrow 0} \frac{3}{\sqrt{3u + 1} + 1}$ |
| (c) $\lim_{x \rightarrow -2} \frac{x^3 + 2x^2 - 1}{5 - 3x}$ | (m) $\lim_{s \rightarrow 1} \frac{1 - s^3}{2 - \sqrt{s^2 + 3}}$ |
| (d) $\lim_{x \rightarrow -1} \frac{x^4 + x^2 - 1}{2x + 1}$ | (n) $\lim_{t \rightarrow -3} \frac{t^2 - 9}{2t^2 + 7t + 3}$ |
| (e) $\lim_{y \rightarrow 2} \sqrt{4y^2 - 3}$ | (o) $\lim_{h \rightarrow 0} \frac{(-5 + h)^2 - 25}{h}$ |
| (f) $\lim_{x \rightarrow 1} \frac{x^2 - 7x + 10}{x - 2}$ | (p) $\lim_{x \rightarrow -2} \frac{x + 2}{x^3 + 8}$ |
| (g) $\lim_{x \rightarrow -1} \frac{x + 1}{x^2 - x - 2}$ | (q) $\lim_{x \rightarrow 2} \frac{\sqrt{4x + 1} - 3}{x - 2}$ |
| (h) $\lim_{x \rightarrow -2} \frac{x^2 - 2x - 8}{x^2 - 4a - 12}$ | (r) $\lim_{x \rightarrow 3} \frac{\frac{1}{3} - \frac{1}{3}}{x - 3}$ |
| (i) $\lim_{x \rightarrow 1} \frac{x - 1}{x^3 - 1}$ | (s) $\lim_{t \rightarrow 0} \frac{\sqrt{1 + t} - \sqrt{1 - t}}{t}$ |
| (j) $\lim_{x \rightarrow -2} \left(1 + \frac{2}{x}\right) \left(\frac{5}{4 - x^2}\right)$ | (t) $\lim_{x \rightarrow -4} \frac{\sqrt{x^2 + 9} - 5}{x + 4}$ |